SUPPLEMENTARY DIGITAL MATERIAL 1

Supplementary Table I.—Selected studies and their main results

Reference (First Author, Year)	Study Design	Sample Size	Conclusions (focus on impact on follow-up care)	PP Sequelae
AlMutiri AM *et al.* 2021 [23]	Case report	1	Neurological complications from spinal cord injury can occur	Devastating spinal cord injury case due to cervical spine fracture-dislocation
Bein T *et al.* 2004 [24]	Prospective randomized trial	59 CPP=25 ICPP=27	• Use of CPP should be preferred • Incidence of PU or face oedema in CPP-group did not differ from ICPP	• Complications: - Stage 1 PU (4/59; 7.7%) - Stage 2 PU (3/59; 5.8%) - Face oedema (N.=36/59; 69.2%)
Bellinghausen AL *et al.* 2020 [25]	Case report	2	Providing “Fed back” on complications into the ICU where patients were treated to improve the quality of care that patients receive	Meralgia paresthetica
Binda F *et al.* 2021 [17]	Cross-sectional study	63	Obesity is not an obstacle for the use of PP	• Prevalence of PU was 42.9% (95%-CI: 30.6–55.1) whereas it was 30.2% (95%-CI: 18.8–41.5) when considering only PP-related PU

				• Patients with PU showed a higher level of correlation (ρ=0.47, P=0.042) between days of mechanical ventilation and PP-time, compared to patients without PU (ρ=0.29, P=0.052) • Localizations of PU (total of 54): 59.2% (32/54) head, 35.2% (19/54) sacrum, genitals or heels, 5.6% (3/54) other
Brugliera L *et al.* 2021 [26]	Case series	7	Healthcare workers should take into consideration the risk factors for PP–related plexopathy and nerve damage	• Case 1: Bilateral axillary, suprascapular nerves' lesion, and bilateral mononeuropathy of ulnar nerve • Case 2: Bilateral ulnar nerve neurapraxia • Case 3: Axonotmesis of the left ulnar nerve • Case 4: Left brachial plexopathy • Case5: Right brachial plexopathy (rostral portion), possible overlap with right musculocutaneous nerve injury; left lumbosacral plexopathy • Case 6: Neurapraxia of the right ulnar nerve and moderate impairment of

				the right median nerve; axonotmesis of the right peroneal nerve • Case 7: Focal right ulnar nerve impairment
Burdet L *et al.* 2001 [27]	Case report	1	Chest support should target maximal relief of pressure on breast implants	Bilateral breast necrosis
Challoner T el al. 2021 [28]	Retrospective cohort study	62	Each prone session had a 3.11 times increased risk of developing PU	PU in 57/62 (91.9%)
Chan MC *et al.* 2007 [29]	Prospective observational clinical study	22 11=PP 11=SP	More position-related complications in PP	• PP-group: - acial swelling (9/11; 81.8%) - Eye swelling (5/11; 45.4%) - Knee swelling (1/11; 9.1%) - Pressure sores (2/11; 18.2%) - Vomiting - Nasogastric pump infusion
Chang LG *et al.* 2021 [30]	Case series	5	Detection and reduction of compressive neuropathies	Unilateral ankle dorsiflexion weakness
Chiumello D *et al.* 2006 [31]	Comparative study	11	Application of thoraco-pelvic supports potentially	• PU: - Thorax - Abdomen - Sacrum/pubis

			increase skin-tissue damage	- Global
Chiumello D *et al.* 2012 [32]	Observational prospective study	26	No differences in pulmonary function or quality of life of ARDS patients between PP and SP	• No significant differences between PP and SP in any of the SF-36 categories • No significant differences between PP and SP concerning SGRQ score
Conlon C *et al.* 2021 [33]	Case report	1	Patients in PP should be evaluated from head to toe for developing PIs	Full-thickness facial PI and buried dentition
Davidson TA *et al.* 1999 [34]	Prospective, matched, parallel cohort study	73	Reduction in physical function	Significant reduction in HRQoL, (physical function) in 7 of 8 SF-36 domains caused by ARDS and its sequelae
Decavel P *et al.* 2021 [35]	Case series	10 PP=9 SP=1	PP ventilation therapy appears to be the major factor for lower cranial nerve palsies	• Cranial nerve palsy (9 had a hypoglossal nerve palsy. Of these 9 patients, 8 had a deficit in 1 other cranial nerve: vagal nerve (7/8), facial nerve (2/8), and glossopharyngeal nerve (2/8)) • Combination of hypoglossal and vagal palsy (5 patients) • 1 patient had an isolated vagus nerve palsy
Douglas IS *et al.* 2021 [36]	Retrospective single-center study	427 PP=61	Brachial plexus palsy and catheter–associated bloodstream	• Generalized, facial and limb oedema was common • DTPI and minor PI were identified on ventral

			infections more common with prolonged PPV	surfaces (chest, abdomen, and groin) in 38 (71.7%) • Wounds on face, chin, nose, and neck are common—25 patients (47.2%) • Extremity weakness in 58 patients (95.1%) with true brachial plexus palsies in five (8.2%)
Dudek NL *et al.* 2002 [37]	Case report	1	Complicating factors included variable levels of oxygenation, malnutrition, anemia, and steroid therapy	Bilateral anterior superior iliac spine PU
Gattinoni L *et al.* 2001 [38]	Multicenter, randomized trial	304 PP=152 SP=152	Number of new or worsening PU are significantly higher in the PP group	PP weight-bearing sites (thorax, cheekbone, iliac crest, breast, and knee) significantly more likely to be affected in PP (70 of 188 sores were at these sites, as compared to 12 of 102 in SP; P<0.001)
Girard R *et al.* 2014 [39]	Randomized controlled trial	466 P P=237 SP=229	• PP was associated with a higher risk of PU than SP • Active prevention for risk-groups is needed	• PU on day 7: (116/204; 57.1%); at ICU discharge: (103/223; 44.4%): - Face (58/197; 29.4%) (41/223; 18.4%) - Back (2/189; 1.1%) (2/214; 0.9%) - Sacrum (40/196; 20.4%) (47/221; 21.3%) - Heel (22/190; 11.6%)

				(22/214; 10.3%) - Back of head (6/191; 3.1%) (1/214; 0.5%) - Anterior part of thorax (35/195;17,9%) (14/219; 6.4%) - Other site (48/190; 25.3%) (36/219; 16.9%)
Gleissman H *et al.* 2021 [40]	Case serie	44	PP may also cause potentially severe complications which should be weighed against the potential benefits	• Complications: - 21 (48%) cases of facial oedema - 18 (41%) cases of pressure sores - 11 (25%) cases of airway complication/ETT obstruction - 1 (2%) case of nerve damage of the arm
Goettler CE *et al.* 2002 [41]	Case report	2	• Extra care must be taken with arm position and motion • Shoulders may not be posteriorly extended	Brachial plexopathy
Grant GP *et al.* 2010 [42]	Pilot study	10	Much attention of ocular perfusion pressure needed	Prolonged PP increases IOP, choroid layer thickness, and optic nerve diameter

Guerin C *et al.* 2004 [43]	Randomized controlled trial	791 PP=413	Some safety concerns associated with PP	Pressure sores, selective intubation, and endotracheal tube obstruction incidences were higher in the PP
Guerin C *et al.* 2018 [2]	Observational Study	735 PPV=101	PP was associated with a low rate of complications, a significant increase in oxygenation and a significant decrease in driving pressure	• Complications: - Endotrachial tube-related complications (2/101) - Hypoxemia (2/101) - Ocular complications (2/101) - Pressure sores (5/101) - Transient increase in intracranial pressure (1/101)
Ibarra G *et al.* 2020 [16]	Case-control study	74	PU are related to the characteristics of the maneuver and the previous nutritional state	• 136 PP pressure sores were recorded • Face was the most affected region (69%) • An increased risk of PU was associated with increased number of days under pronation cycles, and PP maintained for more than 24 h
Jové Ponseti E *et al.* 2017 [44]	Descriptive retrospective transversal study	38	• Relationship between PU and time spent in the PP is not statistically significant, yet a tendency can be observed	• PU (9/38; 25.7%) • Grade 1 (2/9; 22.2%): - Ears - Cheeks - Chin - Back of foot • Grade 2 (6/9; 66.7%): - Cheeks

			• Fewer PU occurred in patients who had received suitable nutrition	- Eyelids - Abdomen - Genitals - Scapular Waist • Grade 3 (0/0) • Grade 4 (1/9; 11.1%) • Facial oedema (8/38; 22.8%) • Conjunctival oedema (5/38; 14.3%)
Juhl CS *et al.* 2016 [45]	Case report	1	Avoid unnecessary pressure when placing a patient with ARDS in PP	Meralgia Paresthetica
Kappel C *et al.* 2020 [46]	Case report	1	Risk factors: obesity, proning (associated with prolonged pressure to the genitals)	Multiple areas of necrotic tissue around the glans of the penis, the superior scrotum, and at the base of the penis
Khan F *et al.* 2019 [47]	Retrospective cohort study	81	Complications of PP were uncommon	13 patients with oedema of the eyes, lips, and tongue
Kiekens C *et al.* 2020 [48]	Interview		Rehabilitation is needed for post-acute phase for patients after severe disease	• Complications: - Dysphagia - Muscle weakness - Critical illness myopathy and neuropathy - Reduced joint mobility - Neck and shoulder pain - Difficulty in verticalization - Impaired balance and gait

				- Limitations in activities of daily living (ADL) - Difficult awakening with long-lasting confusional state and psychological problems
Kim RS *et al.* 2016 [49]	Case report	4	Use of thin soft silicone foam dressings may prevent the development of facial deep tissue injuries	Multiple necrotic facial PU (2/4)
King-Robson J *et al.* 2022 [50]	Case report	2	Plexopathy is a disabling complication of PP	• Case 1: brachial plexus injury • Case 2: plexopathy
Le MQ *et al.* 2019 [6]	Case report	1	Be aware of potential complications of PP	• Large eschar over the sternum extending to the right ribs • Facial wounds, including tissue loss of the lower lip, left nare and left maxillary region • Prominent left deltoid atrophy, weakness of left shoulder abduction and external rotation and left elbow flexion
Léonet S *et al.* 2002 [51]	Verbal questionnaire survey	25	Study identifies considerable reticence among ICU nursing staff	• Complication: - Facial oedema (23) - Decubitus ulcers (14) - Cardiorespiratory arrest (7)

			towards the use of PP	- Extubation (3) - Joint complications (2)
Leuzinger-Dias M *et al.* 2021 [52]	Case report	1	Cooperation between ophthalmologists and intensivists to promote regular eye evaluations	Occult right globe rupture
L´Her E *et al.* 2002 [53]	Prospective observational study	51	• Twelve-hour PP is a safe procedure • The occurrence of PU was directly related to the required PP number of sessions, but not to the mean individual PP duration • It was also correlated to a longer ICU stay, but without any effect on clinical outcome	• Turning adverse events in 448 turning maneuvers (9/224; 4%): - Transient blood pressure decrease in patients under vasoactive drugs (N.=2) - Central venous catheter dislodgement (N.=3) - Hypoxemia related to massive bronchial drainage (N.=1) - Selective endotracheal intubation and barotrauma (N.=1) - Endotracheal extubation (N.=1) - Sinus drain dislodgement (N.=1) • Cutaneous bedsores (23/51; 45%): - Stage 1 or 2 (13/51; 25%) - Stage 3 or 4 (10/51; 20%)

				• Facial oedema (51/51) • Enteral nutrition discontinuation (13/51; 25%) • Vomiting (6/51; 12%) • Increase in gastric residue (7/51; 14%) • Transient intracranial pressure increase (2/51) • Necrotic pressure sores (2/51) • Transient increase in intracranial pressure at the beginning of the prone position session (1/51)
Lucchini A *et al.* 2020 [15]	Single-center, retrospective observational Study	170	• Need of strategies for skin protection • Correct position of patients' face	• Pressure sores: - Face cheekbones (11/170; 6%) (Grade 1: 7/170; Grade 2: 2/170; Grade 3: 2/170) - Face chin (8/170; 5%) (Grade 1: 4/170; Grade 2: 4/170; Grade 2: 4/170) - Thorax (3/170; 2%) (Grade 1: 1/170; Grade 2: 2/170) - Trochanter (1/170; 1%) - Other sites (8/170; 5%) (Grade 1: 2/170; Grade 2: 6/170)

Lucchini A *et al.* 2021 [21]	Retrospective cohort study	96	Extended PP is feasible and might reduce the workload on healthcare workers without significant increase of major PP related complications	• Patients with at least one pressure sore due to PP were 38 (40%) • Patients with PU showed a statistically significative difference in intensive care length of stay, mechanical ventilation days, numbers of PP cycles, total time spent in PP and the use of extended PP, compared to patients without PU • Cheekbones (18%) and chin (10%) were the most affected sites
Malik GR *et al.* 2020 [54]	Case report	83	Physicians must be aware of an increased susceptibility to peripheral nerve injury in severe COVID-19 and refine standard protocols in order to reduce the risk	• 12 patients (14.5%) with peripheral nerve injury • In total, 21 focal peripheral nerve injury sites across these 12 patients and one newly distal symmetric polyneuropathy • Majority of these peripheral nerve injuries occurred in the upper limb (76.2%) • Most frequent sites of injury included the ulnar nerve (six cases, 28.6%), radial nerve (three cases, 14.3%), sciatic nerve (three cases, 14.3%),

				brachial plexus (two cases, 9.5%), and median nerve (two cases, 9.5%)
Mancebo J *et al.* 2006 [55]	Randomized control trial	136	Duration of PP may be an important determinant of its effectiveness, as may be the interval between the onset of ARDS and its application	• Complications: - Oedema (facial, limb, thorax) (14/136; 10%) - Conjunctival hemorrhage (2/136; 1.5%) - Pressure sores (2/136; 1.5%) • Complications due to the maneuver of PP: - Cardiac arrest (1/136; 1%) - Displacement of urinary bladder catheter and nasogastric feeding tube (2/136; 1.5%) - Kinking of endotracheal tube (1/136; 1%) - Kinking of thoracic drain (1/136, 1%)
Marinelli L *et al.* 2020 [56]	Case report	1	Medical personnel should be aware of the risk of meralgia paraesthetica as a disabling condition	• Bilateral muscle weakness of the dorsal extensor muscles of the foot • Meralgia Paraesthetica
Martínez Campayo N *et al.* 2020 [57]	Case report	1	PU-preventive measures should be implemented	Symmetric chest PU

Mayr F *et al.* 2016 [58]	Case report	1	Significant increase in IOP due to PP	Posterior ischemic optic neuropathy (PION) resulting in bilateral vision loss
Miller C *et al.* 2021 [59]	Case series	15	Early identification of peripheral nerve injuries, providing rehabilitation interventions, and referring to specialist services if necessary	• 30 anatomical nerve injuries were recorded: - the most commonly affected nerve was the ulnar nerve (12/30) - followed by the cords of the brachial plexus (10/30) • Neuropathic pain and muscle wasting were identified
Perrillat A *et al.* 2020 [60]	Case report	2	• Intensive care staff must be adequately trained • PP should be supervised and regularly monitored	• Multiple facial PU: - on forehead affecting the eyebrow - cheekbone - cheek - next to the masseter - labial commissure • Patient presented keratitis and a corneal ulcer • Hemifacial swelling cause of masseter myositis
Panchabhai TS *et al.* 2016 [61]	Case report	1	Intensivists need to be aware of potentially modifiable risk factors that could prevent rare complications of PPV (e.g., ischemic optic neuropathy)	Extended PPV resulting in loss of vision from ischemic optic neuropathy

Ramondetta *et al.* 2020 [62]	Case report	1	Dermatologic evaluation is sometimes needed	Presence of multiple ulcerative lesions on the face (chin and nostrils)
Ravella KC *et al.* 2021 [63]	Case report	1	Importance of increased care when positioning patients prone by padding bony prominences, evenly distributing pressure across known sites of peripheral nerve entrapment	Left median nerve mononeuropathy and bilateral meralgia paresthetica
Rodríguez-Huerta MD *et al.* 2021 [64]	Descriptive ambispective study	44	PP requires a nursing protocol to prevent the occurrence of adverse events	• Complications: - Facial oedema (26; 81.3%) - Eye injuries (4; 12.5%) - Accidental device removal, which occurred in only two patients (6.1%) - One case of endotracheal tube obstruction (3.3%) - Other: hematuria, herpes complicating a pressure ulcer, ocular hematoma - • 20 patients (60.6%) developed grade I and II facial PU: - Chin (12; 32.2%) - Forehead (8; 22.9%) - Nose (n6; 17.1%)

				• 5 patients (15.2%) developed one grade I and II PU in the thoracic or abdominal region
Rrapi R *et al.* 2021 [65]	Retrospective cohort study	1216	Taking steps to address predisposing factors is essential in preventing and improving outcomes	• 84 patients (6.9%) with 118 PI • 36 PI (30.5%) occurring on the face
Sahoo JN *et al.* 2019 [66]	Case series	11	PPV improves oxygenation when started early with adequate duration	PU in four patients
Sánchez-Soblechero A *et al.* 2020 [67]	Case report	1	Careful positioning and handling of patients who are in a PP	Upper trunk brachial plexopathy
Saran S *et al.* 2019 [68]	Prospective observational study	25	Significant increase in IOP due to PP among ARDS patients	• Increase of IOP in not dependent eye (NDE) • After turning the patient back to SP, IOP in NDE was found to rapidly decrease
Shearer SC *et al.* 2021 [69]	Retrospective chart review	263 PP=143	There is an urgent need for hospitals to implement preventative measures for these prone facial injuries	• 68 (47.6%) PI to the head and neck regions • Average duration of PP for patients that developed PI was significantly longer • Most commonly affected sites were the cheeks

				(N.=57, 84%), the ears (N.=34, 50%) and the nose (N.=14, 21%)
Taccone P *et al.* 2009 [70]	Multicenter, unblinded, randomized controlled trial	342 PP=168 SP=174	PP does not provide significant survival benefit in patients with ARDS or in subgroups of patients with moderate and severe hypoxemia	• Of 34 patients in the PP (20.2%): - Hemodynamic instability (13 patients) - Massive facial oedema (7 patients) - Malfunction of continuous renal replacement therapy (3 patients) - Other reason (eg, potential dislodgment of a chest/tracheotomy tube, cerebral oedema, massive alveolar hemorrhage) (11 patients)
Terlemez R *et al.* 2021 [71]	Case report	1	Careful clinical examination and electrophysiological studies are crucial for the diagnosis of ICU- related entrapment neuropathies	Peroneal nerve palsy
Trejo-Gabriel-Galan JM *et al.* 2017 [72]	Case report	1	PPV can be infrequent cause of dysphagia	Lower Carnial Nerves Paralysis

Willems MCM *et al.* 1997 [73]	Case report	2	Correct position of patients' arms	• Calcifications of both shoulders on roentgenograms and extensive myositis ossificans • During PPV, bilateral necrosis of the papilla mammae
Zhou L *et al.* 2021 [74]	Case report	2	Neuropathy may be preventable by using egg-crate style foam padding between hip and bed	Lateral femoral cutaneous neuropathy
Zingarelli EM *et al.* 2020 [75]	Case report	1	Be aware of all the COVID-19 related complications	Multiple facial skin lesions und PUs

PP: Prone Position; CPP: Complete Prone Position; ICPP: Incomplete Prone Position; SP: Supine Position; PU: Pressure Ulcer; PI: Pressure Injury; DTPI: Deep Tissue Pressure Injury; SF-36: Short Form-36 Questionnaire; SGRQ: St. George’s Respiratory Questionnaire.

References

1. Guérin C, Reignier J, Richard JC, Beuret P, Gacouin A, Boulain T *et al.* PROSEVA Study Group. Prone positioning in severe acute respiratory distress syndrome. N Engl J Med 2013;368(23):2159-68.
2. Guérin C, Beuret P, Constantin JM, Bellani G, Garcia-Olivares P, Roca O, *et al.* A prospective international observational prevalence study on prone positioning of ARDS patients: the APRONET (ARDS Prone Position Network) study. Intensive Care Med 2018;44(1):22-37.
3. Rawal G, Yadav S, Kumar R. Acute Respiratory Distress Syndrome: An Update and Review. J Transl Int Med 2018;6(2):74-7.
4. Xu Z, Shi L, Wang Y, Zhang J, Huang L, Zhang C, *et al.* Pathological findings of COVID-19 associated with acute respiratory distress syndrome. Lancet Respir Med 2020;8:420–422.
5. Rothan HA, Byrareddy SN. The epidemiology and pathogenesis of coronavirus disease (COVID-19) outbreak. J Autoimmun 2020;109:102433.
6. Le MQ, Rosales R, Shapiro LT, Huang LY. The Down Side of Prone Positioning: The Case of a Coronavirus 2019 Survivor. Am J Phys Med Rehabil 2020;99(10):870-872.
7. Elmer N, Liebl ME, Schwedtke C, Drebinger D, Reißhauer A. A [Acute rehabilitation after COVID-19 infection : A selected case series]. Z Rheumatol. 2022;81(5):386-392. German
8. Koulouras V, Papathanakos G, Papathanasiou A, Nakos G. Efficacy of prone position in acute respiratory distress syndrome patients: A pathophysiology-based review. World J Crit Care Med. 2016;5(2):121-136.
9. Langer T, Brioni M, Guzzardella A, Carlesso E, Cabrini L, Castelli G, *et al.* Prone position in intubated, mechanically ventilated patients with COVID-19: a multi-centric study of more than 1000 patients. Crit. Care 2021;25:128.
10. Kallet RH. A Comprehensive Review of Prone Position in ARDS. Respir Care 2015;60(11):1660-87.
11. Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften (AWMF). S3-Leitlinie – Empfehlungen zur stationären Therapie von Patienten mit COVID-19 [11th Working Group of Scientific Medical Societies (AWMF). S3 guideline - recommendations for inpatient treatment of patients with COVID-19]. AWMF Online; 2022 [Internet]. Available from: https://www.awmf.org/uploads/tx_szleitlinien/113-001LGl_S3_Empfehlungen-zur-stationaeren-Therapie-von-Patienten-mit-COVID-19_2021-10_1.pdf [cited 2022, Nov 17].

12. Bellani G, Laffey JG, Pham T, Fan E, Brochard L, Esteban A, *et al.* Epidemiology, Patterns of Care, and Mortality for Patients With Acute Respiratory Distress Syndrome in Intensive Care Units in 50 Countries. JAMA 2016;315(8):788-800.
13. Scholten EL, Beitler JR, Prisk GK, Malhotra A. Treatment of ARDS with prone positioning. Chest 2017;151: 215–224.
14. Xu Q, Wang T, Qin X, Jie Y, Zha L, Lu W. Early awake prone position combined with high-flow nasal oxygen therapy in severe COVID-19: a case series. Crit Care 2020;24(1):250.
15. Lucchini A, Bambi S, Mattiussi E, Elli S, Villa L, Bondi H, *et al.* Prone Position in Acute Respiratory Distress Syndrome Patients: A Retrospective Analysis of Complications. Dimens Crit Care Nurs. 2020;39(1):39-46.
16. Ibarra G, Rivera A, Fernandez-Ibarburu B, Lorca-García C, Garcia-Ruano A. Prone position pressure sores in the COVID-19 pandemic: The Madrid experience. J Plast Reconstr Aesthet Surg 2021;74(9):2141-2148.
17. Binda F, Galazzi A, Marelli F, Gambazza S, Villa L, Vinci E, *et al.* Complications of prone positioning in patients with COVID-19: A cross-sectional study. Intensive Crit Care Nurs 2021;67:103088.
18. Team V, Team L, Jones A, Teede H, Weller CD. Pressure Injury Prevention in COVID-19 Patients With Acute Respiratory Distress Syndrome. Front Med (Lausanne) 2021;7:558696.
19. Dirkes S, Dickinson S, Havey R, O'brien D. Prone positioning: is it safe and effective? Crit Care Nurs Q 2012;35(1):64-75.
20. Mora-Arteaga JA, Bernal-Ramírez OJ, Rodríguez SJ. The effects of prone position ventilation in patients with acute respiratory distress syndrome. A systematic review and metaanalysis. Med Intensiva 2015;39(6):359-72.
21. Lucchini A, Russotto V, Barreca N, Villa M, Casartelli G, Marcolin Y, *et al.* Short and long-term complications due to standard and extended prone position cycles in CoViD-19 patients. Intensive Crit Care Nurs 2022;69:103158.
22. PRISMA statement. PRISMA; 2020 [Internet]. Available from: www.prisma-statement.org [cited 2022, Nov 17].
23. AlMutiri AM, Alsulaimani S, Sabbagh AJ, Bajunaid KM, Tashkandi WA, Baeesa SS. Cervical Spinal Cord Injury During Prone Position Ventilation in the COVID-19 Pandemic. Cureus 2021;13(10):e18958.
24. Bein T, Sabel K, Scherer A, Papp-Jambor C, Hekler M, Dubb R, *et al.* Comparison of incomplete (135 degrees) and complete prone position (180 degrees) in patients with acute

respiratory distress syndrome. Results of a prospective, randomised trial. Anaesthesist 2004;53(11):1054-60.

25. Bellinghausen AL, LaBuzetta JN, Chu F, Novelli F, Rodelo AR, Owens RL. Lessons from an ICU recovery clinic: two cases of meralgia paresthetica after prone positioning to treat COVID-19-associated ARDS and modification of unit practices. Crit Care 2020;24(1):580.
26. Brugliera L, Filippi M, Del Carro U, Butera C, Bianchi F, Castellazzi P, *et al.* Nerve Compression Injuries After Prolonged Prone Position Ventilation in Patients With SARS-CoV-2: A Case Series. Arch Phys Med Rehabil 2021;102(3):359-362.
27. Burdet L, Liaudet L, Schaller MD, Broccard AF. Bilateral breast necrosis after prone position ventilation. Intensive Care Med 2001;27:1435.
28. Challoner T, Vesel T, Dosanjh A, Kok K. The risk of pressure ulcers in a proned COVID population. Surgeon. 2022;20(4):e144-e148.
29. Chan MC, Hsu JY, Liu HH, Lee YL, Pong SC, Chang LY, *et al.* Effects of prone position on inflammatory markers in patients with ARDS due to community-acquired pneumonia. J Formos Med Assoc 2007;106(9):708-16.
30. Chang LG, Zar S, Seidel B, Kurra A, Gitkind A. COVID-19 Proned Ventilation and Its Possible Association With Foot Drop: A Case Series. Cureus. 2021;13(4):e14374.
31. Chiumello D, Cressoni M, Racagni M, Landi L, Li Bassi G, Polli F, *et al.* Effects of thoraco-pelvic supports during prone position in patients with acute lung injury/acute respiratory distress syndrome: a physiological study. Crit Care 2006;10(3):R87.
32. Chiumello D, Taccone P, Berto V, Marino A, Migliara G, Lazzerini M, *et al.* Long-term outcomes in survivors of acute respiratory distress syndrome ventilated in supine or prone position. Intensive Care Med 2012;38(2):221-9.
33. Conlon C, Slovacek C, Jalalabadi F, Winocour S, Olorunnipa O. Full-Thickness Facial Pressure Injury and Buried Dentition from Prone Positioning in a Patient with COVID-19. Adv Skin Wound Care 2021;34(8):1-3.
34. Davidson TA, Caldwell ES, Curtis JR, Hudson LD, Steinberg KP. Reduced quality of life in survivors of acute respiratory distress syndrome compared with critically ill control patients. Jama 1999;281(4):354-60.
35. Decavel P, Nahmias O, Petit C, Tatu L. Lower Cranial Nerve Palsies in the COVID-19 Pandemic: A 10-Case Series of Intensive Care Unit Patients. Eur Neurol 2021;23:1-4.
36. Douglas IS, Rosenthal CA, Swanson DD, Hiller T, Oakes J, Bach J, *et al.* Safety and outcomes of prolonged usual care prone position mechanical ventilation to treat acute coronavirus disease 2019 hypoxemic respiratory failure*. Crit Care Med 2021;49:490–502.

37. Dudek NL, Buenger UR, Trudel G. Bilateral anterior superior iliac spine pressure ulcers: a case report. Arch Phys Med Rehabil 2002;83(10):1459-61.
38. Gattinoni L, Tognoni G, Pesenti A, Taccone P, Mascheroni D, Labarta V, *et al.* Prone-Supine Study Group. Effect of prone positioning on the survival of patients with acute respiratory failure. N Engl J Med 2001;345(8):568-73.
39. Girard R, Baboi L, Ayzac L, Richard JC, Guérin C. The impact of patient positioning on pressure ulcers in patients with severe ARDS: results from a multicentre randomised controlled trial on prone positioning. Intensive Care Med 2014;40(3):397-403.
40. Gleissman H, Forsgren A, Andersson E, Lindqvist E, Lipka Falck A, Cronhjort M, *et al.* Prone positioning in mechanically ventilated patients with severe acute respiratory distress syndrome and coronavirus disease 2019. Acta Anaesthesiol Scand 2021;65(3):360-363.
41. Goettler CE, Pryor JP, Reilly PM. Brachial plexopathy after prone positioning. Crit Care 2002;6(6):540-2.
42. Grant GP, Szirth BC, Bennett HL, Huang SS, Thaker RS, Heary RF, *et al.* Effects of prone and reverse trendelenburg positioning on ocular parameters. Anesthesiology 2010;112:57–65.
43. Guerin C, Gaillard S, Lemasson S, Ayzac L, Girard R, Beuret P, *et al.* Effects of systematic prone positioning in hypoxemic acute respiratory failure: a randomized controlled trial. JAMA 2004;292(19):2379-87.
44. Jové Ponseti E, Villarrasa Millán A, Ortiz Chinchilla D. Analysis of complications of prone position in acute respiratory distress syndrome: quality standard, incidence and related factors. Enferm Intensiva 2017;28(3):125-34.
45. Juhl CS, Ballegaard M, Bestle MH, Tfelt-Hansen P. Meralgia Paresthetica after Prone Positioning Ventilation in the Intensive Care Unit. Case Rep Crit Care 2016;2016:7263201.
46. Kappel C, Piticaru J, Jones G, Goucher G, Cheon P, Fischer M, *et al.* A case of possible Fournier's gangrene associated with proning in COVID-19 ARDS. Can J Anaesth 2020;67(11):1697-1698.
47. Khan F, Fistler CR, Mixell J, Caplan R, Vest MT. Community Experience With Acute Respiratory Distress Syndrome in the Prone Position. Crit Care Explor 2019;1(12):e0068.
48. Kiekens C, Boldrini P, Andreoli A, Avesani R, Gamna F, Grandi M, *et al.* Rehabilitation and respiratory management in the acute and early post-acute phase. "Instant paper from the field" on rehabilitation answers to the Covid-19 emergency. Eur J Phys Rehabil Med 2020;56(3):323-326.
49. Kim RS, Mullins K. Preventing Facial Pressure Ulcers in Acute Respiratory Distress Syndrome (ARDS). J Wound Ostomy Continence Nurs 2016;43(4):427-9.

50. King-Robson J, Bates E, Sokolov E, Hadden RDM. Prone position plexopathy: an avoidable complication of prone positioning for COVID-19 pneumonitis? BMJ Case Rep 2022;4;15(1):e243798.
51. Léonet S, Fontaine C, Moraine JJ, Vincent JL. Prone positioning in acute respiratory failure: survey of Belgian ICU nurses. Intensive Care Med 2002;.28(5):576-80.
52. Leuzinger-Dias M, Lima-Fontes M, Oliveira-Ferreira C, Camisa E, Sousa C, Rocha-Sousa A, *et al.* Prone Positioning Covid-19 Patients: A Double-Edged Sword-A Case Report of a Devastating Ocular Complication. Ophthalmol Ther 2021;10(3):691-697.
53. L'Her E, Renault A, Oger E, Robaux MA, Boles JM. A prospective survey of early 12-h prone positioning effects in patients with the acute respiratory distress syndrome. Intensive Care Med 2002;28(5):570-5.
54. Malik GR, Wolfe AR, Soriano R, Rydberg L, Wolfe LF, Deshmukh S, *et al.* Injury-prone: peripheral nerve injuries associated with prone positioning for COVID-19-related acute respiratory distress syndrome. Br J Anaesth 2020;125(6):e478-e480.
55. Mancebo J, Fernández R, Blanch L, Rialp G, Gordo F, Ferrer M, *et al.* A multicenter trial of prolonged prone ventilation in severe acute respiratory distress syndrome. Am J Respir Crit Care Med 2006;173(11):1233-9.
56. Marinelli L, Mori L, Avanti C, Cotellessa F, Fabbri S, Schenone C, *et al.* Meralgia Paraesthetica after Prone Position Ventilation in a Patient with COVID-19. Eur J Case Rep Intern Med 2020;7(12):002039.
57. Martínez Campayo N, Bugallo Sanz JI, Mosquera Fajardo I. Symmetric chest pressure ulcers, consequence of prone position ventilation in a patient with COVID-19. J Eur Acad Dermatol Venereol 2020;34(11):e672-e673.
58. Mayr F, Asher N, Scanlan K, Butcher B. Bilateral blindness after prone positioning for acute respiratory distress syndrome. Critical Care Medicine 2016;44(12):563.
59. Miller C, O'Sullivan J, Jeffrey J, Power D. Brachial Plexus Neuropathies During the COVID-19 Pandemic: A Retrospective Case Series of 15 Patients in Critical Care Phys Ther 2021;101(1):pzaa191.
60. Perrillat A, Foletti JM, Lacagne AS, Guyot L, Graillon N. Facial pressure ulcers in COVID-19 patients undergoing prone positioning: How to prevent an underestimated epidemic?. J Stomatol Oral Maxillofac Surg 2020;121(4):442-444.
61. Panchabhai TS, Bandyopadhyay D, Kapoor A, Akindipe O, Lane C, Krishnan S. Acute ischemic optic neuropathy with extended prone position ventilation in a lung transplant recipient. Int J Crit Illn Inj Sci 2016;6(1):45-7.

62. Ramondetta A, Ribero S, Costi S, Dapavo P. Pression-induced facial ulcers by prone position for COVID-19 mechanical ventilation. Dermatol Ther 2020;33(4):e13748.
63. Ravella KC, Redondo ML, Mejia A, Gonzalez MH. Median Nerve Mononeuropathy and Meralgia Paresthetica After Prone Positioning in a Patient with COVID-19 ARDS: A Case Report. JBJS Case Connect 2021;11(3).
64. Rodríguez-Huerta MD, Díez-Fernández A, Rodríguez-Alonso MJ, Robles-González M, Martín-Rodríguez M, González-García A. Nursing care and prevalence of adverse events in prone position: Characteristics of mechanically ventilated patients with severe SARS-CoV-2 pulmonary infection. Nurs Crit Care. 2022;27(4):493-500.
65. Rrapi R, Chand S, Lo JA, Gabel CK, Song S, Holcomb Z, *et al.* The significance of pressure injuries and purpura in COVID-19 patients hospitalized at a large urban academic medical center: A retrospective cohort study. J Am Acad Dermatol. 2021;85(2):462-464.
66. Sahoo JN, Gurjar M, Mohanty K, Majhi K, Sradhanjali G. Prone ventilation in H1N1 virus-associated severe acute respiratory distress syndrome: A case series. Int J Crit Illn Inj Sci 2019;9(4):182-186.
67. Sánchez-Soblechero A, García CA, Sáez Ansotegui A, Fernández-Lorente J, Catalina-Álvarez I, Grandas F, *et al.* Upper trunk brachial plexopathy as a consequence of prone positioning due to SARS-CoV-2 acute respiratory distress syndrome. Muscle Nerve 2020;62(5):E76-E78.
68. Saran S, Gurjar M, Kanaujia V, Ghosh PS, Gupta A, Mishra P, *et al.* Effect of Prone Positioning on Intraocular Pressure in Patients With Acute Respiratory Distress Syndrome. Crit Care Med 2019;47(9):e761-e6.
69. Shearer SC, Parsa KM, Newark A, Peesay T, Walsh AR, Fernandez S, *et al.* Facial Pressure Injuries from Prone Positioning in the COVID-19 Era. Laryngoscope 2021;131(7):E2139-E2142.
70. Taccone P, Pesenti A, Latini R, Polli F, Vagginelli F, Mietto C, *et al.* Prone-Supine II Study Group. Prone positioning in patients with moderate and severe acute respiratory distress syndrome: a randomized controlled trial. JAMA 2009;302(18):1977-84.
71. Terlemez R, Özekli Mısırlıoğlu T, Palamar D, Okutan D, Akgün K. Bilateral foot drop after COVID-19-related acute respiratory distress syndrome: A case report. Turk J Phys Med Rehabil 2021;67(3):378-381.
72. Trejo-Gabriel-Galan JM, Perea-Rodriguez ME, Aicua-Rapun I, Martinez-Barrio E. Lower Cranial Nerves Paralysis Following Prone-Position Mechanical Ventilation. Crit Care Med 2017;45(8):e865-e6.

73. Willems MCM, Voets AJ, Welten RJTJ. Two unusual complications of prone-dependency in severe ARDS. Intensive Care Med 1998;24:276-281.
74. Zhou L, Siao P. Lateral femoral cutaneous neuropathy caused by prone positioning to treat COVID-19-associated acute respiratory distress syndrome. Muscle Nerve 2021;63(6):E50-E52.
75. Zingarelli EM, Ghiglione M, Pesce M, Orejuela I, Scarrone S, Panizza R. Facial Pressure Ulcers in a COVID-19 50-year-old Female Intubated Patient. Indian J Plast Surg 2020;53(1):144-146.
76. Hahn J-M. Dekubitus [Decubitus]. In: Hahn JM, editor. Checkliste Innere Medizin [Internal medicine checklist]. Leipzig: Georg Thieme Verlag; 2018.
77. Edsberg LE, Black JM, Goldberg M, McNichol L, Moore L, Sieggreen M. Revised National Pressure Ulcer Advisory Panel Pressure Injury Staging System: Revised Pressure Injury Staging System. J Wound Ostomy Continence Nurs 2016;43(6):585-97.
78. Poage C, Roth C, Scott B. Peroneal nerve palsy: Evaluation and management. J Am Acad Orthop Surg 2016;24:1–10.
79. Gehlen W, Delank HW. Meralgia paraesthetica [Paraesthetic meralgia]. In: Gehlen W, Delank HW, editors. Neurologie [Neurology]. Leipzig: Georg Thieme Verlag; 2010.
80. Majmundar N, Shastri D, Assina R, Goldstein I. Meralgia paresthetica following prone position in posterior lumbar spinal surgery: Case series and review of the literature. Interdisciplinary Neurosurgery 2016;8.
81. Short Form – 36. In: Brooks NP, Strayer AL, editors. Spine Surgery in an Aging Population. First Edition. Leipzig: Thieme Medical Publishers, Inc.; 2019.
82. Herridge MS, Cheung AM, Tansey CM, Matte-Martyn A, Diaz-Granados N, Al-Saidi F, *et al.* One-year outcomes in survivors of the acute respiratory distress syndrome. N Engl J Med 2003;348(8):683-93.
83. Sanghi P, Malik M, Hossain IT, Manzouri B. Ocular complications in the prone position in the critical care setting: The COVID-19 pandemic. Journal of Intensive Care Medicine 2021;36(3):361–372.
84. Lang GE. Subkonjunktivale Hämorrhagien (Hyposphagma) [Subconjunctival hemorrhages (hyposphagma)]. In: Suttorp N, Möckel M, Siegmund B, Dietel M, editors. Harrisons Innere Medizin [Harrison's Internal Medicine]. Leipzig: ABW Verlag; 2020.
85. Kwee M, Ho Y-H, Rozen W. The Prone Position During Surgery and its Complications: A Systematic Review and Evidence-Based Guidelines. International surgery 2015;100:292-303.

86. Hearne BJ, Hearne EG, Montgomery H, Lightman SL. Eye care in the intensive care unit. J Intensive Care Soc 2018;19(4):345-50.
87. Machiele R, Motlagh M, Patel BC. Intraocular Pressure. Treasure Island, FL: StatPearls Publishing; 2020.
88. Striebel HW. Intrakranieller Druck [Intracranial pressure]. In: Striebel HW, editor. Anästhesie Intensivmedizin Notfallmedizin [Anesthesia Intensive Care Medicine Emergency Medicine]. Leipzig: Georg Thieme Verlag; 2020.
89. Braga IA, Brito CS, Filho AD, Filho PPG, Ribas RM. Pressure ulcer as a reservoir of multiresistant Gram-negative bacilli: risk factors for colonization and development of bacteremia. Braz J Infect Dis 2017;21:171–5.
90. World Health Organization. Rehabilitation in health systems. Geneva: World Health Organization; 2017.